

MSWIPE WAREHOUSE INWARD AND REVERSE INWARD OPERATIONS

A Supply Chain Case Study on MPOS Device Lifecycle, ERP Controls, Quality Checks, RMA, Allocation, and Dispatch

Field	Details
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Focus Area	Supply Chain Operations, Warehouse Management, Reverse Logistics
Company Context	Mswipe Technologies Private Limited
Process Scope	Fresh inward, Refurbished Inward, QC, RMA, Allocation, Dispatch, Installation Support
Source Basis	Mswipe Logistics SOP, July 2025, combined with portfolio-level process analysis

This case study is written as a portfolio-safe operational analysis. It explains the end-to-end warehouse flow at a high level while still showing detailed supply chain thinking, process mapping, control points, documentation requirements, exception handling, and improvement opportunities.

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1. EXECUTIVE SUMMARY

This case study analyzes the warehouse inward and reverse inward operations used to manage MPOS devices across the Mswipe device lifecycle. The process begins before the warehouse physically receives product. It starts with demand planning, procurement, purchase order approval, supplier documentation, customs clearance, and the pre-upload of device-level information into ERP. Once the devices reach the HO-Juinagar warehouse, the logistics team performs physical scanning, validates device IDs against finance-uploaded records, updates ERP, and moves matched units into the correct store location.

The main operational strength of the process is that it combines physical controls with system controls. A device does not simply enter inventory because a box arrived. The device must be individually scanned, matched to expected records, and reflected in ERP and the flat file. This creates traceability at the device ID level and reduces the risk of inventory loss, duplicate records, or incorrect allocation.

The reverse inward process is more complex because returned devices may come from FSEs, consignees, direct merchants, or remote service cases. These devices can have different statuses such as installation completed, pending installation, pending activation, pending deinstallation, or free device. The store team must coordinate ticket closure before the device can be moved to a free device status and brought back into the warehouse flow. After that, the device moves through store verification, QC, possible RMA repair, another round of QC, and eventual allocation or dispatch.

From a supply chain point of view, this operation is not just warehouse receiving. It is a closed-loop asset management system. Mswipe is handling both forward logistics and reverse logistics for serial-numbered devices that have operational, financial, and customer-facing value. The inward process protects inventory accuracy. The reverse inward process recovers value from returned devices. QC protects customer experience. RMA reduces unnecessary replacement cost. Allocation and dispatch convert available inventory into field-ready deployment.

Case Study Theme	What It Shows
Device-level inventory control	Every MPOS device is tracked by device ID through scanning, ERP updates, flat file updates, transfers, QC, allocation, and dispatch.
Cross-functional coordination	Treasury, finance, logistics, store, QC, RMA, operations, helpdesk, accounts, FSEs, and vendors all touch the lifecycle.
Forward and reverse logistics	Fresh devices enter from suppliers, while faulty or used devices return from the field and are recovered through QC and RMA.
Process controls	Short inventory, excess inventory, QC failures, damaged merchant devices, and ERP transfers all have defined escalation paths.
Portfolio value	The project demonstrates warehouse operations, process mapping, ERP workflow, asset traceability, reverse logistics, and operational risk management.

2. BUSINESS CONTEXT AND OPERATING ENVIRONMENT

Mswipe operates in the payments technology space, where MPOS devices and related hardware are deployed to merchants. These devices are not generic warehouse items. They are field assets that must be traceable, functional, properly configured, and connected to merchant installation or replacement tickets. Because the devices eventually support payment acceptance, the warehouse process has to protect both operational speed and control accuracy.

The logistics environment includes multiple movement types. New devices are imported and inwarded into the central warehouse. Devices are dispatched to FSEs or consignees for installation. Devices may return from the field as faulty, unused, deinstalled, damaged, or replacement inventory. Some devices are repaired and reused. Some may need supplier replacement. Some are directly dispatched to remote merchants. Each

movement changes the device status and location, so ERP accuracy becomes a core operational requirement.

The warehouse is connected to a broader operating system. Procurement begins with stock availability and sales forecasts. Finance and treasury handle PO and supplier documents. Logistics manages physical receipt and scanning. Store teams control warehouse location movement. QC checks firmware, slots, application loading, ROM, and entry-level screens. RMA repairs defective units. Operations uses Odoo pending reports and ticket data to determine where devices should go. Dispatch uses courier integrations and local handover controls. FSEs use the F2A2 application to install, map, replace, and close field tickets.

This environment creates a real supply chain challenge: one device can move through many hands, many systems, and many statuses before it becomes productive at a merchant location. The case study therefore focuses on how Mswipe controls the physical and digital flow of devices from inward to field deployment, and from field return back into usable inventory.

3. SCOPE OF THE CASE STUDY

The scope of this case study is the operational flow around warehouse inward and reverse inward. It starts at procurement handoff and ends when a device becomes available for allocation, dispatch, installation, or replacement. The analysis also includes related processes that affect inward quality, such as QC, RMA, ERP stock transfer, allocation, dispatch, installation, and device upgrade support.

Included Area	Explanation
Fresh inward	New imported MPOS devices received at HO-Juinagar after supplier dispatch, documentation sharing, customs clearance, ERP pre-upload, and physical scanning.
Reverse inward	Refurbished, faulty, non-working, or returned devices received from FSEs, consignees, or direct merchants.
QC	Quality checks performed after inward or repair to validate device readiness before allocation.
RMA	Repair and maintenance workflow for devices requiring part replacement, supplier replacement, or additional testing support.
ERP and flat file controls	Location updates, stock transfers, scanning results, batch-wise stock checks, and F2A2 visibility.
Allocation and dispatch	Device assignment based on requirements, tickets, pending reports, courier dispatch, and local FSE collection.
Installation and upgrade linkage	How F2A2 field activity closes the loop after device dispatch or replacement.

The case study does not attempt to disclose exact internal screens, employee-level responsibilities, or confidential commercial terms. Instead, it explains the process in a way that is useful for a supply chain portfolio and appropriate for a business audience.

4. KEY STAKEHOLDERS, SYSTEMS, AND DOCUMENTS

A major feature of the Mswipe logistics process is that it is cross-functional. The warehouse team cannot complete the process alone because inventory accuracy depends on upstream documents, system uploads, quality confirmations, and downstream tickets. Each stakeholder owns a different control point.

Stakeholder	Main Role in the Process
MD/CFO	Provides quantity and rate details based on inventory availability and sales forecast inputs.
Senior Treasury Manager and Treasury Department	Prepares purchase order details, coordinates supplier documentation, and supports the import purchase flow.
Finance Team	Uploads device details into ERP, receives discrepancy reports, and coordinates with OEM or suppliers for shortages, excess quantities, or replacements.
Logistics Team	Receives devices, scans device IDs, performs ERP inward updates, coordinates stock movement, manages courier tracking, and supports dispatch.
Store Team	Controls physical warehouse receipt, verifies returned devices, transfers devices to QC, receives QC-cleared inventory, and supports local handover.
QC Team	Checks device functionality and readiness using defined quality parameters and marks inspected units with label status.
RMA Team	Assesses faulty units, replaces parts when available, requests procurement for unavailable parts, and coordinates supplier replacement when needed.
Operations Team	Maintains device requirement lists, shares service request details, uses Odoo reports, and supports allocation decisions.
Helpdesk Team	Raises tickets for installation, replacement, upgrade, damaged device cases, and merchant-reported issues.
FSEs and Consignees	Receive devices, perform merchant installation or replacement, collect damaged units when required, and close tickets in F2A2.
Accounts Team	Maps receipts, posts vouchers, supports billing, booking, payment, and charges linked to upgrades or repair exceptions.

System or Document	Operational Purpose
ERP	Main system for device stock upload, inward, stock transfer, courier API dispatch, batch-wise stock lookup, and location changes.
Flat file	Operational tracking file that reflects device status, location, and movement updates outside or alongside ERP.
Odoo Pending Ops Report	Daily operations report showing pending installation devices categorized by location.
F2A2 application	Field app used by FSEs to view assigned devices, receive transfers, map devices to merchants, upload photos, and close tickets.
Inward register	Excel-based register for returned or refurbished devices, including device details, consignee/FSE details, receipt date, courier details, box number, and status.
Purchase order and proforma invoice	Commercial and documentation basis for supplier procurement and inward validation.
Packing list and device details	Supplier-shared records used to support finance upload and warehouse verification.
Inspection report	QC record maintained in Excel to document device inspection or verification status.
Courier docket / airway bill	Shipment document generated through courier integration and attached to the device box for dispatch.
E-way bill	Generated when shipment value exceeds the applicable threshold listed in the SOP.

5. END-TO-END MPOS DEVICE LIFECYCLE

The MPOS device lifecycle can be viewed as a closed loop. A device may begin as new imported inventory, move into store stock, pass QC, get allocated, dispatched, installed at a merchant, later return because of

service or replacement, go through reverse inward, be repaired or refurbished, pass QC again, and re-enter allocation stock. This closed loop is the core of the case study.

HIGH-LEVEL FORWARD FLOW

Step 1	Planning and purchase requirement based on inventory position and sales forecast.
Step 2	PO preparation, CFO approval, supplier proforma invoice, shipping and customs clearance.
Step 3	Supplier shares invoice, packing list, and device details with finance or treasury.
Step 4	Finance uploads expected device details into ERP through bulk upload.
Step 5	HO-Juinagar logistics team receives goods and physically scans each device ID.
Step 6	ERP validates scanned device IDs against finance-uploaded records.
Step 7	Matched devices move into store inventory and are updated in ERP and flat file.
Step 8	One sample unit may be sent for functionality testing, then returned to store.
Step 9	Devices move to QC, receive inspection status, and return to store if passed.
Step 10	Available devices are allocated and dispatched based on tickets, reports, or remote merchant requirements.

HIGH-LEVEL REVERSE FLOW

Step 1	Device returns from FSE, consignee, merchant, or field activity as faulty, non-working, returned, or pending status inventory.
Step 2	Inward team records device in inward register with device, box, consignee/FSE, receipt date, courier, and current status information.
Step 3	Store team checks whether device status is free or whether ticket closure is required before movement.
Step 4	Once cleared, ERP and flat file movement shift the device from FSE/consignee location to inward location and then store location.
Step 5	Store verifies device condition against the inward register and sends device to QC.
Step 6	QC checks quality parameters and either returns device to store or forwards it to RMA.
Step 7	RMA repairs device when possible, coordinates for parts or supplier replacement when needed, and returns repaired device to QC.
Step 8	QC retests repaired device and returns passed inventory to store for allocation and dispatch.

6. FRESH INWARD PROCESS DEEP DIVE

Fresh inward refers to new MPOS devices imported into the business and received at the HO-Juinagar warehouse. The fresh inward process is important because it establishes the first official inventory record for new devices. If this step is inaccurate, every later movement becomes unreliable. A device that is not properly inwarded may not be visible for allocation. A device that is inwarded incorrectly may show the wrong location, wrong status, or wrong availability. For this reason, the process uses both documentation matching and physical scanning.

6.1 PRE-RECEIPT DOCUMENTATION

The process begins when goods listed in the proforma invoice are handed over to the shipping agent. The supplier shares invoice copy, packing list, and device details with the finance or treasury team. This creates the expected receipt record before the shipment physically arrives at the warehouse. Finance then uploads the device details into ERP through a bulk upload. This step converts supplier-side information into company-side system records.

From an operations standpoint, this is a preventive control. The warehouse should not be the first place where the company learns what devices are arriving. By uploading expected device IDs in advance, the warehouse scanning process becomes a match check rather than a manual entry exercise. This reduces data entry errors and prevents unmatched devices from being accepted into stock without investigation.

6.2 PHYSICAL RECEIPT AT HO-JUINAGAR

Once customs clearance and receipt at the port are completed, the devices arrive at the HO-Juinagar warehouse. The logistics team physically receives the goods and scans each device ID. The scanning activity is not only a receiving activity. It is also a reconciliation step. ERP allows only matched entries to proceed because scanned device data is cross-verified against the records uploaded by finance.

This device-level control is important because MPOS inventory cannot be managed only by carton quantity. The warehouse needs to know which exact devices are available, where they are located, and whether they are ready for deployment. Capturing the device ID in ERP creates asset traceability from the first warehouse touchpoint.

6.3 HANDLING MATCHED, SHORT, AND EXCESS INVENTORY

If scanned devices match the uploaded ERP records, the logistics team processes the inward update and the devices move to the store location. If there is a discrepancy, the process separates shortage and excess scenarios. In a shortage, the finance department informs the OEM or supplier and keeps the shortage open until the missing devices arrive in a later lot or batch. In an excess case, the finance department informs the OEM or supplier, and the excess stock is held separately until instructions are received.

The shortage and excess treatment is a strong inventory control because it prevents the warehouse from casually absorbing supplier errors. Short inventory is not ignored simply because the shipment is mostly correct. Excess inventory is not immediately mixed with available stock. Both cases are escalated through finance, which is the right function to coordinate supplier-level commercial resolution.

Fresh Inward Control Point	Purpose	Risk Reduced
Supplier documents shared before arrival	Creates expected receipt record before physical scanning	Missing records, unclear shipment content, manual guesswork
Finance bulk upload into ERP	Loads device details for validation	Unmatched device IDs, duplicate records, wrong initial master data

Physical scanning by logistics	Captures actual received device IDs	Carton-level receiving errors, untracked devices
ERP cross-verification	Allows matched entries to proceed	Acceptance of unapproved or unknown devices
Flat file update	Reflects operational movement/status	Mismatch between system and operational tracking
Short inventory escalation	Keeps missing quantity open until resolved	Supplier shortage hidden inside warehouse process
Excess inventory segregation	Keeps extra units separate until instructions	Unauthorized stock mixing or financial mismatch

6.4 SAMPLE FUNCTIONALITY TESTING

After fresh inward, one unit is sent to the Product Team at the Marathon Corporate Office in Mumbai for functionality testing. Once the team confirms that the device is working properly, the unit is returned to the store. The stock movement to the testing team and back to store follows an ERP stock transfer process, similar to an inter-batch stock transfer.

This step acts like a sample validation gate. It verifies that newly inwarded devices are not only present in inventory but also operationally functional. The transfer is still recorded in ERP, which means the device is not lost while outside the store location for testing.

6.5 MOVEMENT TO QC

After the sample testing confirmation, devices move to the quality department for QC. The store manager performs the ERP transfer and the device moves to the QC department grid. This creates a formal handoff between store and QC. Because the ERP transfer mirrors the physical transfer, the business can identify whether a device is in store, in QC, in RMA, dispatched, or assigned to an FSE.

The fresh inward process therefore ends only when devices have been received, validated, recorded, tested where required, and moved into the quality flow. At that point, the inventory becomes a controlled asset ready for inspection and eventual allocation.

7. ERP AND FLAT FILE CONTROL LOGIC

ERP is the system backbone of the inward process. The uploaded device ID flows into ERP after scanning, and once a batch of scanning is complete, the list of device IDs is populated on screen. The logistics team processes the upload, after which the device location moves to store. Updated location can be checked through inventory, item stocks, batch-wise stock, and flat file views.

The flat file is also important because it provides an operational view of device status and location. In many warehouse environments, ERP is the official record while operational teams still rely on files or reports for daily execution. The key control is that the ERP and flat file must remain aligned. When a device moves from FSE to inward, inward to store, store to QC, QC to RMA, RMA back to QC, or store to FSE, both the system location and the operational tracker must reflect that movement.

Movement Type	ERP/Tracker Impact	Operational Meaning
Fresh inward scan	Device ID is populated and processed in ERP	New device becomes visible as received stock.
Store location update	Device location moves into store location	Inventory becomes warehouse-controlled.
Store to QC transfer	Stock transfer entry shifts device to QC grid	Device is under inspection, not available for dispatch.
QC to store transfer	Stock transfer returns passed unit to store	Device is ready for allocation.

QC to RMA transfer	Stock transfer shifts failed unit to repair team	Device is blocked until repair and retest.
FSE/consignee to inward	Returned device is moved back into inward flow	Reverse logistics process begins.
Store to FSE via F2A2	Device moves to FSE bucket after OTP verification	Field agent owns next action and device visibility updates in app.
Courier dispatch	Docket and airway bill are generated and attached to shipment	Device exits warehouse through courier process.

A strong device-level ERP process also creates accountability. If a device is assigned to an FSE, it should be visible in the F2A2 application. If a device is in QC, it should not be treated as available. If a damaged returned device is in RMA, it should not be dispatched until the repair and QC retest are complete. This is the operating logic that keeps the physical warehouse and digital records aligned.

8. QUALITY CHECK PROCESS

The QC process is the main quality gate between warehouse stock and customer-facing deployment. The store team informs the QC team about stock transfers through email communication, and the transfer is also processed in ERP. Once QC receives the devices, the store team receives an acknowledgment email confirming receipt. This creates a documented handoff and reduces the risk of unclear ownership.

QC checks devices according to parameters set by the quality department. These include firmware, slot working, application uploading, ROM check, entry-level screen, and other checks listed in the quality plan. The inspection or verification is recorded in an Excel inspection report, and a QC logo or label is affixed to identify inspected status.

The label system is simple but effective. Freshly inward devices receive green labels after QC. Refurbished devices receive yellow labels after QC. Devices received from RMA post-repair receive orange labels. This visual management system helps warehouse staff quickly distinguish inventory condition and process history without opening a system screen every time.

QC Label	Applied To	Process Meaning
Green	Freshly inward devices after QC completion	New inventory passed QC and is ready for store return and allocation.
Yellow	Refurbished devices after QC completion	Returned or refurbished inventory passed QC and can re-enter usable stock.
Orange	Devices received from RMA after repair	Repaired inventory passed through maintenance and must be clearly distinguishable.

If a device passes QC and meets defined quality parameters, it is returned to store for allocation and dispatch through a standard stock transfer entry. If a device requires repair after quality checks, QC forwards it to RMA through the standard stock transfer process, and the transfer details are communicated by email and acknowledged by RMA. This prevents failed devices from being accidentally mixed into available inventory.

The QC process is especially important in a payments device operation because a device failure in the field can create merchant dissatisfaction, repeat service visits, delayed onboarding, and additional logistics cost. By catching issues before dispatch, QC acts as both a quality control function and a cost avoidance function.

9. REPAIRS AND MAINTENANCE PROCESS

The RMA team handles devices that fail QC or require repair after returning from the field. Once a device reaches RMA, the team evaluates the device condition and determines whether any parts require replacement. If the necessary part is available, it is replaced. For minor issues, RMA coordinates with the testing team for support. This creates an internal repair path instead of immediately replacing the entire device.

If a part is irreplaceable, RMA informs the finance team, which coordinates with the supplier for full device replacement. Replacement details are maintained in Excel by device ID. If a required part is unavailable, procurement is initiated through the standard process. An indent is raised in Coconet, supported by a consumption report and HOD approval. Procurement then raises a purchase order, and parts are received at HO according to agreed delivery terms. After receipt, those parts are used for device repairs.

Once RMA completes repair, the device does not go directly to store or dispatch. It returns to QC for another round of testing. Only if the repaired device passes QC does it return to store for allocation and dispatch. This is a critical control because repair completion does not automatically equal deployment readiness.

RMA REPAIR LOOP

Step 1	QC identifies device issue and transfers device to RMA through ERP stock transfer.
Step 2	RMA evaluates physical and functional condition of the device.
Step 3	If parts are available, RMA replaces the required parts.
Step 4	If part issue is minor, RMA coordinates with testing team if additional support is needed.
Step 5	If part is irreplaceable, RMA informs finance for supplier or OEM replacement coordination.
Step 6	If part is unavailable, indent, consumption report, HOD approval, PO, and parts receipt are initiated.
Step 7	Repaired device is transferred back to QC through ERP.
Step 8	QC retests the device and either returns it to store or sends it back for further repair.

The RMA process supports reverse logistics value recovery. Without an RMA process, every faulty return could become scrap or full replacement. With RMA, the company can recover usable assets, reduce replacement cost, extend device life, and increase available inventory for future installations or replacements.

10. REVERSE INWARD AND REFURBISHED DEVICE PROCESS

Reverse inward is the return flow for refurbished, faulty, non-working, or returned devices received from Mswipe FSEs and direct merchants. This process is more complicated than fresh inward because returned devices have history. They may be linked to tickets, merchants, deinstallation activities, pending activations, or field service assignments. Before the device can be treated as reusable inventory, its status must be checked and cleared.

10.1 INWARD REGISTER AND INITIAL CAPTURE

Returned devices received at the store location are recorded in both the Excel-based inward register and ERP. The inward register captures device information, consignee name, FSE details, date of receipt, and courier information when applicable. It also includes a summary of device name and box number, which is shared with the store team. Another column reflects the device status as per the flat file.

The inward register is the first control document in the reverse flow. It bridges the gap between physical receipt and digital status. A returned device cannot be managed properly unless the team knows which device came back, who sent it, when it was received, which box it was in, and what status it currently holds.

10.2 STATUS VALIDATION BEFORE FREE DEVICE MOVEMENT

The inward register includes statuses such as Installation Completed, Pending for Installation, Pending for Activation, Pending for Deinstallation, and Free Devices. The free device status matters because it indicates the device is no longer blocked by a field ticket or merchant status. For all statuses other than Free Device, the store team coordinates with the respective FSEs or consignees to ensure ticket closure. Once the ticket is closed, the store team informs the inward department by email so the device transfer from FSE to inward location can begin.

This step prevents devices from being returned into warehouse stock while still operationally tied to an unresolved ticket. It protects against duplicate allocation, inaccurate field records, and premature reuse of devices that are still marked as active or pending.

Returned Device Status	Operational Interpretation	Required Action
Installation Completed	Device may still be linked to completed installation history	Confirm whether it can be released as free inventory.
Pending for Installation	Device is linked to an installation activity that is not closed	Coordinate ticket closure or status update before inward movement.
Pending for Activation	Device may be installed but not fully activated	Resolve activation or ticket status before reuse.
Pending for Deinstallation	Device is expected to be removed but process is not closed	Ensure deinstallation closure before free status.
Free Device	Device is not blocked by an open operational status	Move through inward, store, QC, and potential allocation flow.

10.3 MOVEMENT FROM FSE OR CONSIGNEE TO INWARD AND STORE

Once the device becomes free, ERP entry is performed using the same stock transfer logic. The device movement is updated from FSE or consignee to inward location in ERP, and the same change is reflected in the flat file. From the inward location, devices move to the store or warehouse location. This is the point where the returned unit physically and digitally re-enters the warehouse process.

At store location, free devices are verified against the inward register. The store team checks the condition of each device and then sends it to the quality department. The movement from store to QC is again done

through stock transfer logic, and the flat file location shifts from FSE to store. This repeat use of stock transfer rules keeps the process standardized across fresh, reverse, QC, and RMA movements.

10.4 QC AND RMA ROUTING FOR RETURNED DEVICES

Once QC checks the returned device, the device is either labeled quality done and found okay or sent to RMA for repairs. If it is found okay, it is returned to store. If not, RMA checks the condition and replaces parts when needed. In the standard returned device repair flow, the cost is not charged to the merchant. RMA then returns the device to QC, and ERP records the transfer.

This creates a controlled refurbishing loop. The company is not simply accepting returned devices as usable. It is validating status, verifying physical receipt, checking condition, performing QC, repairing where needed, retesting, and only then making the device available for future allocation.

11. EXCEPTION HANDLING FOR DAMAGED DEVICES

The SOP includes a specific exception case for damaged devices reported by merchants. If the merchant reports damage through the Helpdesk Team, the HD Team raises a TBK request. An FSE then visits the merchant location to inspect the device condition. The RMA team assesses the damage and provides an estimated repair cost to the FSE, who communicates the cost to the merchant.

If the merchant agrees to pay, the FSE shares merchant details with RMA. A Pay-By-Link is sent to the merchant registered mobile number. Once payment is completed, the merchant shares payment confirmation with the FSE, and the FSE forwards it to the RMA team through WhatsApp or Mattermost. After RMA confirmation, the FSE collects the damaged device from the merchant and submits it to HO for repair through the standard inward process: inward, store, QC, and RMA.

This exception process is important because damaged devices are not treated the same as normal faulty returns. The merchant payment confirmation becomes part of the control flow before collection and repair. It prevents confusion over who is responsible for repair cost and ensures that the damaged unit still returns through the same inward and repair chain once collected.

DAMAGED DEVICE EXCEPTION FLOW

Step 1	Merchant reports damaged device through Helpdesk Team.
Step 2	Helpdesk raises TBK request or ticket.
Step 3	FSE visits merchant location and inspects device condition.
Step 4	RMA estimates repair cost and shares it with FSE.
Step 5	FSE communicates cost to merchant.
Step 6	If merchant agrees, Pay-By-Link is sent to registered mobile number.
Step 7	Merchant completes payment and shares confirmation with FSE.
Step 8	FSE forwards confirmation to RMA through approved communication channel.
Step 9	After confirmation, FSE collects damaged device.
Step 10	Device is submitted to HO and follows inward, store, QC, and RMA process.

12. ALLOCATION AND DISPATCH PROCESS

Once devices are ready in store, the next step is allocation. The store department receives device requirements from multiple sources. These include Ops Pending Reports from Odoo, service requests for replacements, and direct merchant requirements in remote areas where FSE support may not be available. A consolidated list of device requirements is maintained by operations using a Google Sheet. The allocation team reviews requirements and assigns device request tickets in Odoo to the nearest store, consignee, or available FSE.

Allocation is a planning and control step. It connects available inventory with field demand. If allocation is inaccurate, the wrong device may be sent to the wrong location, an FSE may not receive stock on time, or a merchant installation could be delayed. By mapping each requirement in allocation files, the team creates tracking and accountability.

Requirement Source	What It Means	Supply Chain Action
Ops Pending Report from Odoo	Daily report of devices pending installation by location	Allocate available devices to pending installation tickets.
Service request	Replacement need shared by operations through ticketing flow	Assign replacement device and prepare dispatch or FSE transfer.
Direct merchant remote location	Merchant location without FSE support nearby	Dispatch directly from HO using courier or coordinated channel.
Operations Google Sheet	Consolidated demand list maintained by operations	Use as allocation planning file and dispatch reference.
Nearest store/consignee/FSE mapping	Location-based assignment logic	Reduce travel, dispatch time, and unnecessary transfers.

12.1 COURIER DISPATCH

Dispatch and docket generation are handled through API integration with courier partners such as Delhivery and DTDC. In ERP, the team opens the logistics module, courier API, and buffer stock. From and to details are selected for the device allocation. The from location is typically Juinagar, and the destination location is selected based on allocation. The device ID is entered, and device details including model flow onto the screen. Price and weight are auto-populated. The team selects delivery partner, enters box count and dimensions, and generates the docket or airway bill. The airway bill is fixed on the device box before dispatch.

For shipments whose total value exceeds the threshold listed in the SOP, an E-way Bill is generated based on the docket number. Courier bill tracking is performed by logistics and then sent to accounts for billing, booking, and payment. This connects physical shipment, compliance documentation, and financial settlement.

12.2 PAPER ROLL DISPATCH

Paper roll dispatch follows a more manual process. Merchants place a request with the Helpdesk Team. The dispatch team logs into the courier partner website, generates a token number, and attaches that token to the paper roll box before dispatch. This difference shows that not every warehouse item follows the same digital workflow. High-value device dispatch is more integrated, while consumable dispatch may remain partly manual.

12.3 LOCAL DISPATCH THROUGH FSE

For local dispatches, the FSE visits the HO or regional store to collect devices. The store or regional team hands over the devices after a system-authorized transfer. The transfer is initiated through the F2A2 application, where the FSE receives a one-time password. The FSE shares the OTP with the stores or regional

team, and once verified, the devices move from store inventory to the FSE bucket. The flat file is updated as well.

The OTP handover is a strong control for local dispatch. It confirms that the person collecting the device is the intended FSE and that the system transfer happens at the same time as the physical handover. This reduces the risk of undocumented stock movement.

13. INSTALLATION AND UPGRADE LINKAGE

The logistics process does not end when the warehouse dispatches the device. The installation process closes the forward loop. The dispatch team allocates devices based on installation TBKs raised and refers to the Pending Ops Report from Odoo. Once the consignee or FSE receives the device, allocated devices become visible in the F2A2 app.

The SOP defines installation TAT by tier: Tier-1 locations are T+1, Tier-2 locations are T+2, and Tier-3 locations are T+4. This provides a service-level expectation based on location category. The FSE uses F2A2 to view job lists, access ticket details, map the physical device ID to the merchant, map VPA or QR where required, run a test transaction, train the merchant, capture shop and device photos, complete required forms, and submit the ticket for closure.

Location Category	Installation TAT	Operational Meaning
Tier-1	T+1	Fastest expected completion due to stronger accessibility or service coverage.
Tier-2	T+2	Moderate completion window.
Tier-3	T+4	Longer completion window, likely due to distance or service complexity.

Device upgrades follow a similar replacement and dispatch logic. Upgrade requests may come from a merchant, FSE, or Relationship Manager assigned by pin code. Additional charges are communicated to the merchant according to the product-wise upgrade list. Depending on the request path, merchant consent may be captured through agreement copy or recorded line. Payment may be made through NEFT, Pay-By-Link, or PPS. Accounts maps the receipt and posts the voucher. Once the ticket moves to Ops for service, logistics arranges and dispatches the requested device. The FSE then receives the device in F2A2, replaces the old device, maps the new device ID, performs a test transaction, uploads required photos, and closes the ticket.

This linkage matters because it shows the warehouse is connected to revenue and service processes. A device upgrade is not just a dispatch event. It includes merchant consent, payment collection, accounts posting, ticket movement, logistics arrangement, FSE replacement, device mapping, test transaction, and field documentation.

14. RISK AND CONTROL MATRIX

The following risk and control matrix summarizes the main operational risks in the inward and reverse inward process. This is the kind of view that makes the case study stronger for a portfolio because it shows not only what happens, but why each control exists.

Process Area	Risk	Control in Place	Suggested Extra Control
Fresh inward	Devices received but not matched to expected records	ERP cross-verifies scanning against finance-uploaded device details	Daily unmatched device exception report.

Fresh inward	Short inventory not escalated	Finance informs OEM and shortage stays open until next lot or batch	Aging tracker for shortages by supplier and batch.
Fresh inward	Excess inventory mixed into usable stock	Excess stock is set aside until supplier instructions	Separate excess inventory quarantine location in ERP.
QC	Failed devices accidentally dispatched	QC transfers failed devices to RMA and labels passed inventory	System block on dispatch for non-QC-passed status.
RMA	Repair completed but device still faulty	Repaired devices return to QC for retesting	Repair history by device ID with repeat failure flag.
Reverse inward	Returned device still tied to open ticket	Store coordinates ticket closure before free device status	Automated status validation before inward transfer.
Dispatch	Wrong device sent to wrong location	Device ID, location, courier docket, and allocation file mapping	Barcode scan before airway bill print.
Local handover	FSE collects device without system authorization	F2A2 OTP verification moves stock to FSE bucket	Handover audit log with collector and verifier timestamp.
ERP/flat file	Mismatch between system and tracker	Flat file updated with movements	Daily ERP versus flat file reconciliation.
Damaged device exception	Repair cost dispute with merchant	Pay-By-Link payment confirmation before collection	Link payment reference directly to TBK/ticket and device ID.

15. KPI DASHBOARD RECOMMENDATIONS

A case study is stronger when it connects process execution to measurable performance. The SOP defines process steps, but a management dashboard would help track whether the process is working effectively. The following KPIs could be used to monitor inward, reverse inward, QC, RMA, allocation, and dispatch performance.

KPI	Formula or Definition	Why It Matters
Fresh inward cycle time	Time from warehouse receipt to ERP store location update	Measures receiving speed and scanning productivity.
Scan match rate	Matched scanned device IDs divided by total scanned device IDs	Measures accuracy of supplier records and finance upload.
Short inventory count	Number of expected devices not received by batch	Tracks supplier or shipment shortages.
Excess inventory count	Number of unexpected devices received by batch	Tracks supplier/shipment overages and quarantine needs.
QC pass rate	Devices passed by QC divided by devices inspected	Measures incoming or refurbished quality.
RMA turnaround time	Time from QC to RMA transfer until QC return after repair	Measures repair process speed.
Repeat repair rate	Devices returning to RMA more than once divided by repaired devices	Identifies recurring device issues.
Reverse inward clearance time	Time from returned device receipt to free device/store/QC status	Measures reverse logistics efficiency.
Allocation accuracy	Correct device-location assignments divided by total assignments	Measures planning quality.
Dispatch TAT	Time from allocation approval to airway bill/docket generation	Measures outbound speed.
Installation TAT adherence	Installations completed within tier TAT divided by total installations	Measures field execution linkage.
ERP-flat file variance	Count of devices where ERP and flat file location/status differ	Measures master data reliability.

These KPIs could be shown in a simple dashboard with sections for inbound volume, QC status, RMA aging, reverse inward aging, dispatch pending, and installation TAT. The most important dashboard metric would be ERP-flat file variance because poor data alignment affects every other downstream activity.

16. PROCESS IMPROVEMENT OPPORTUNITIES

The existing process already has strong controls, especially at the device ID, ERP transfer, QC, RMA, and F2A2 handover levels. The improvement opportunities below are not criticisms of the process. They are portfolio-style recommendations that show how the operation could be made more scalable, measurable, and audit-friendly.

16.1 CREATE A SINGLE DEVICE LIFECYCLE DASHBOARD

A device lifecycle dashboard could show the exact count of devices by status: in transit, received, scanned, store, QC, RMA, repaired, dispatch-ready, allocated, courier dispatched, FSE bucket, installed, returned, and damaged exception. This would help managers see bottlenecks without switching between ERP, flat files, Odoo, F2A2, Google Sheets, and email trails.

16.2 AUTOMATE REVERSE INWARD STATUS VALIDATION

The reverse inward process depends heavily on whether the returned device is a Free Device or still tied to another status. A control could automatically block inward movement unless the linked ticket is closed or approved for exception handling. This would reduce manual coordination with FSEs and consignees.

16.3 ADD AGING BUCKETS FOR EXCEPTIONS

Short inventory, excess inventory, QC failed devices, RMA pending devices, and reverse inward pending statuses should be aged by days open. A simple 0-2 day, 3-7 day, 8-15 day, and 15+ day view would help identify stale exceptions that need management attention.

16.4 IMPROVE COURIER AND CONSUMABLE DISPATCH STANDARDIZATION

Device dispatch is API-integrated with courier partners, while paper roll dispatch appears more manual. If paper roll volumes are significant, standardizing token generation and tracking through a lightweight dashboard could reduce manual errors and improve dispatch visibility.

16.5 BUILD DEVICE-LEVEL REPAIR HISTORY

RMA replacement details are maintained device ID wise in Excel. A structured repair history would help identify recurring failure patterns by model, batch, supplier, or issue type. This could support supplier quality discussions and better spare parts planning.

16.6 USE BARCODE VALIDATION AT KEY HANDOFFS

The process already uses scanning in fresh inward. Similar scan validations could be applied before QC handoff, RMA handoff, dispatch packing, and FSE handover. This would reduce the risk of wrong-device movement and create timestamped accountability.

Improvement	Expected Benefit	Priority
Device lifecycle dashboard	Better visibility across all device statuses and bottlenecks	High
Automated reverse inward status check	Fewer devices re-entering stock while tickets are open	High
Exception aging tracker	Faster resolution of shortage, excess, QC, RMA, and status delays	High
Repair history by device ID	Better failure analysis and spare parts planning	Medium
Barcode scan at handoffs	More accurate movement control and audit trail	Medium
Standardize paper roll dispatch tracking	Improved consumable visibility and less manual follow-up	Medium

17. GLOSSARY

Term	Meaning in This Case Study
MPOS	Mobile point of sale device or payment terminal handled by the logistics operation.
Fresh inward	Receiving new imported devices into warehouse inventory.
Reverse inward	Receiving returned, faulty, refurbished, non-working, or field-returned devices back into the warehouse flow.
ERP	Enterprise system used to upload stock, perform stock transfers, update locations, and support dispatch.
Flat file	Operational tracking file used to maintain device location and status visibility.
FSE	Field Service Executive or Engineer who receives devices, visits merchants, performs installation or replacement, and closes field tasks.
QC	Quality check team responsible for inspecting device functionality and readiness.
RMA	Repairs and Maintenance team responsible for evaluating and repairing defective devices.

TBK	Ticket or service request used to manage installation, service, damage, or replacement activity.
Odoo Pending Ops Report	Operations report listing pending device installation requirements by location.
F2A2	Field application used by FSEs to receive devices, map devices to merchants, upload photos, and close tickets.
Docket / Airway bill	Courier shipment reference generated for dispatch and attached to the device box.
E-way Bill	Shipment document generated when shipment value exceeds the threshold defined in the SOP.
Free Device	Returned device status showing the device is not blocked by open tickets or pending field actions.
VPA / QR mapping	Mapping process for products where a QR or payment identifier must be assigned to the merchant.

18. CONCLUSION

The Mswipe warehouse inward and reverse inward process is a detailed example of how modern supply chain operations manage serial-numbered technology assets. The process is not only about receiving boxes or shipping devices. It is about controlling the full lifecycle of each device through documentation, ERP validation, scanning, flat file updates, quality checks, repair loops, allocation logic, courier dispatch, field app visibility, and installation closure.

Fresh inward creates the first controlled inventory record for imported devices. Reverse inward recovers value from field-returned devices and prevents usable hardware from being lost, duplicated, or prematurely redeployed. QC protects merchant experience by ensuring devices are functional before dispatch. RMA reduces cost by repairing devices before replacement. Allocation and dispatch convert available inventory into field execution. F2A2 closes the loop by helping FSEs map devices to merchants, complete test transactions, capture documentation, and close tickets.

The biggest takeaway is that Mswipe logistics operates as a closed-loop supply chain. Devices flow forward from supplier to warehouse to field, and they also flow backward from merchant or FSE to inward, QC, RMA, and back to store. Managing this loop requires accuracy, speed, and accountability. A strong portfolio project can use this case study to demonstrate understanding of warehouse management, reverse logistics, ERP workflows, quality control, service operations, and supply chain process improvement.

PORTFOLIO DESCRIPTION

This case study analyzes the warehouse inward and reverse inward operations behind Mswipe MPOS device management. It covers fresh import receiving, ERP-based device ID scanning, stock transfer controls, flat file updates, QC labeling, RMA repair loops, refurbished device inwarding, damaged-device exceptions, allocation, courier dispatch, local FSE handover, installation linkage, upgrade replacement flow, risk controls, KPIs, and improvement opportunities. The project demonstrates supply chain process mapping, warehouse operations, reverse logistics, inventory control, ERP workflow, and operational analysis.